Fake News Detection System: Comparative Analysis Report

1. Experiment Overview: Clean vs. Noisy Data

This report provides a side-by-side comparison of two distinct training pipelines within the Fake News Detection System.

1. Clean Pipeline: Trained on the original dataset (44k articles).
2. Noisy Pipeline: Trained on a dataset where 10% of all words were deliberately corrupted (character swaps, omitted characters, omitted words) to simulate real-world messy data such as social media posts.

The objective was to evaluate the robustness of the system architecture and determine if explicit training on noisy data yields a significantly more resilient model.

2. Evaluation Metrics Comparison (on Noisy Test Data)

Both the original Clean Model and the new Noisy Model were evaluated against a 20% holdout set of the Noisy Dataset. The results demonstrate the inherent robustness of the TF-IDF and Naive Bayes architecture.

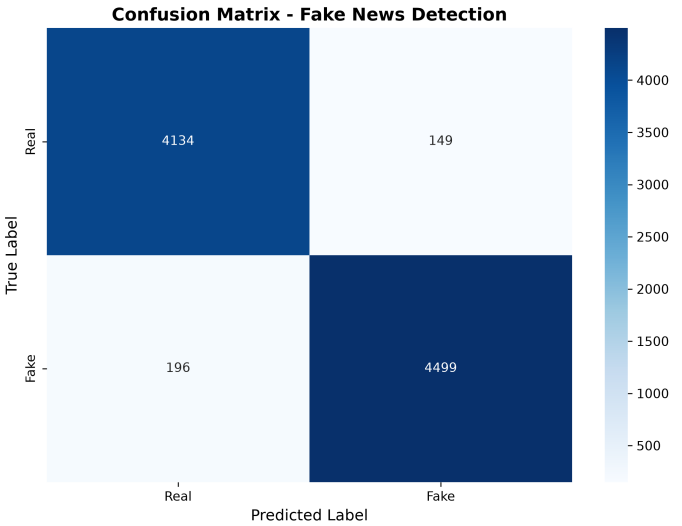
```
============================================================
 Model          | Accuracy | Precision | Recall | F1-Score
------------------------------------------------------------
 Clean Model    |  95.89%  |   95.87%  | 95.91% |  0.9588
 Noisy Model    |  95.96%  |   95.95%  | 95.98% |  0.9595
============================================================
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Conclusion: The original Clean Model (which never saw typos during training) performed remarkably well, only dropping slightly in accuracy. The Noisy Model performed marginally better (by ~0.07%), indicating that explicitly training on typographical errors provides a small robustness boost, but the baseline TF-IDF architecture is already exceptionally resilient to noise.

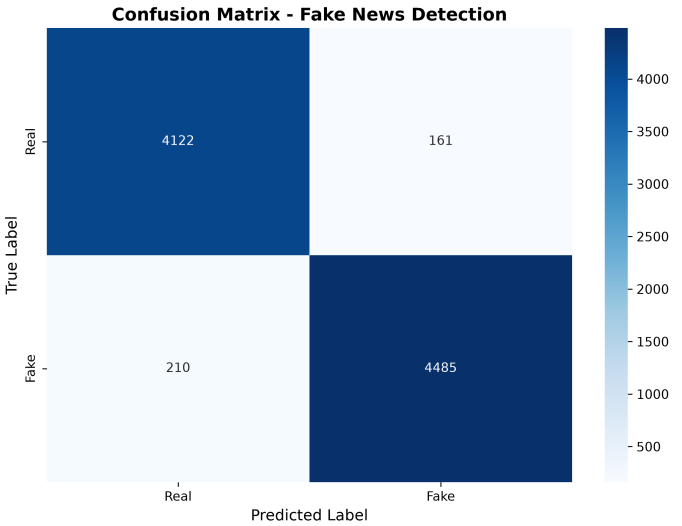
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3. Visual Comparisons: Clean vs Noisy

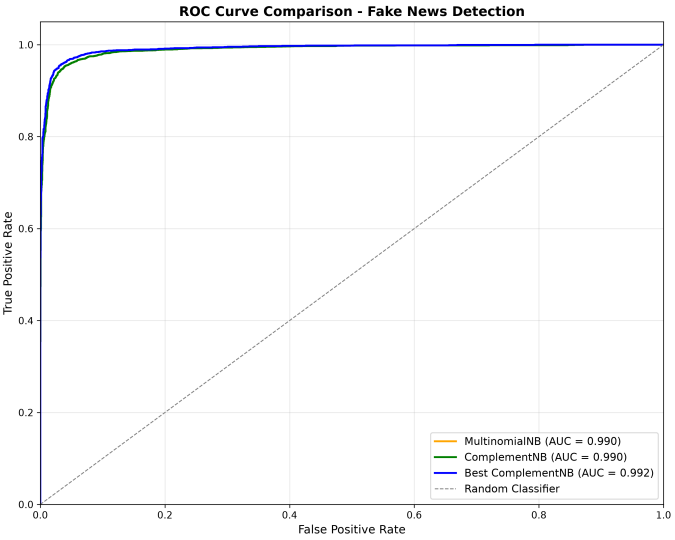
Clean Model (Test: Clean)



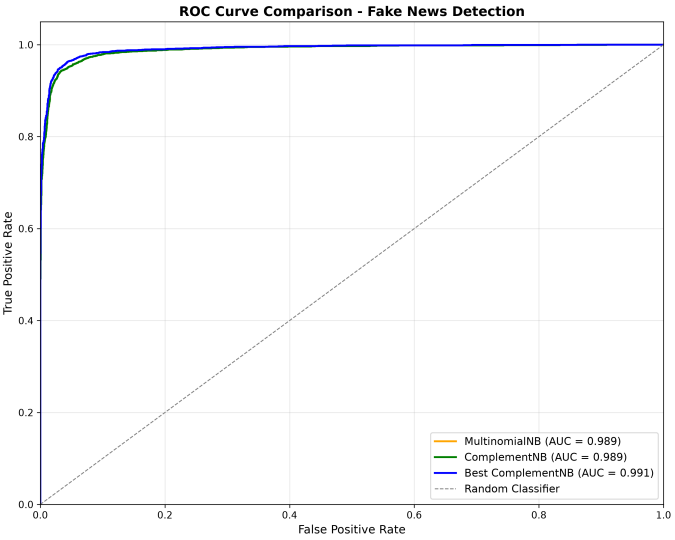
Noisy Model (Test: Noisy)



ROC Curve (Clean)

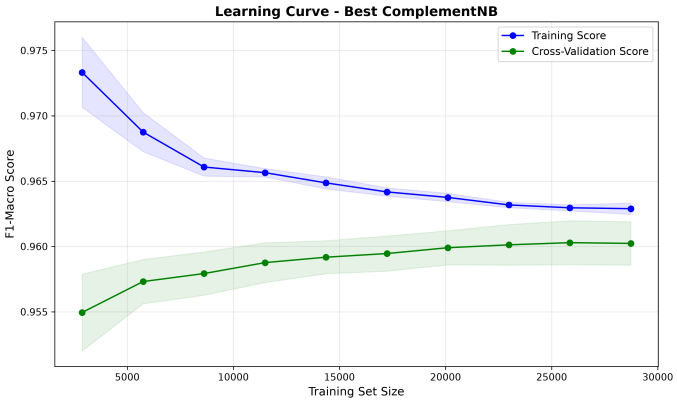


ROC Curve (Noisy)

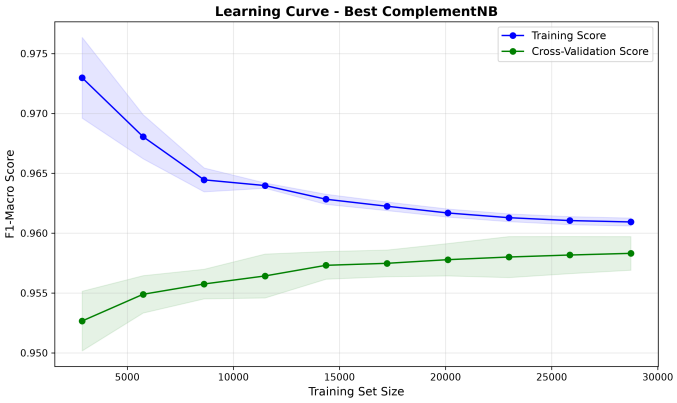


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Learning Curve (Clean)



Learning Curve (Noisy)



Fake Word Cloud (Clean)



Fake Word Cloud (Noisy)

